



LANGUAGE IMPLEMENTERS • CONTRIBUTORS

LANGUAGE CORE MANUAL

ANTLR4 grammar, ASDL AST, builder, unparser, type system,
linter

Open-source Pine Script evaluation framework



CONTENTS

01	Overview
02	Grammar Antlr4
03	Asdl Schema
04	Builder
05	Helper Api
06	Unparser
07	Visitor Transformer
08	Type System
09	Linters
10	Error Model

OVERVIEW

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title: "Language Core" description: "ANTLR4 grammar, ASDL AST, builder, visitors, unparser, type system, linter, and error model for PYNE."

LANGUAGE CORE

The language core is the front half of the evaluate contract: source text becomes a typed tree; the tree can be walked, rewritten, dumped, and unparsed without ever entering the bar loop.

Abstract

PYNE treats Pine Script as a **formal pipeline**, not a host plugin. Lexing and parsing are ANTLR4-driven. The concrete syntax tree is lowered by a hand-written visitor (`PinescriptASTBuilder`) into ASDL-generated dataclasses. Public helpers in `helper.py` mirror Python's `ast` module: `parse`, `unparse`, `dump`, `walk`,



`literal_eval` , plus a process-local sha256 LRU parse cache (`clear_parse_cache` , `parse_cache_info`). Static tools (linter, type registry, LSP) consume the same tree. Evaluation is out of scope here — see [Runtime](#).

Hard rule: edit grammar only under `resource/` . Paths under `generated/` are regenerated artifacts and must not be hand-edited.

Conceptual model

```
flowchart TB
    SRC["Source string / file"]
    LEX["PinescriptLexer"]

    + LexerBase INDENT/DEDENT"]
    TOK["CommonTokenStream"]
    PAR["PinescriptParser"]
    PT["Parse tree contexts"]
    BLD["PinescriptASTBuilder"]
    ANN["Comment → annotations"]
    AST["ASDL AST

Script | Expression"]
    VIS["NodeVisitor / Transformer"]
    UNP["NodeUnparser"]
    LIN["PineLinter"]
    TYP["TypeRegistry"]
    EVAL["Evaluator / LSP / API"]

    SRC --> LEX --> TOK --> PAR --> PT --> BLD --> AST
    TOK -. -> |COMMENT tokens| ANN
    BLD --> ANN
    ANN --> AST
    AST --> VIS
    AST --> UNP
    AST --> LIN
    AST --> TYP
    AST --> EVAL
```

Stage	Role	Primary path
Grammar	Concrete syntax	<code>src/pynescrypt/ast/grammar/antlr4/resource/*.g4</code>
ASDL	Algebraic node shape	<code>src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl</code>
Builder	CST → AST	<code>src/pynescrypt/ast/builder.py</code>
Helper	Public parse/unparse API	<code>src/pynescrypt/ast/helper.py</code>
Nodes	Re-export generated types	<code>src/pynescrypt/ast/node.py</code>

Interface surface

Consumers almost always enter through `pynescrypt.ast.helper` :



```
from pynescrypt.ast.helper import parse, unparse, dump, walk, literal_eval

tree = parse('//@version=6\nindicator("x")\nplot(close)')
print(dump(tree, indent=2))
print(unparse(tree))
```

Successful `parse` results are cached by `sha256(source) + mode` (default ON; `PYNE_PARSE_CACHE=0` disables).
Treat returned trees as **read-only**.

Modes:

mode	Entry rule	Root node
"exec" (default)	<code>start_script</code>	<code>Script</code>
"eval"	<code>start_expression</code>	<code>Expression</code>

Full page inventory:

Page	Topic
Grammar (ANTLR4)	Resource vs generated; typed names; <code>=</code> vs <code>:=</code>
ASDL schema	Algebraic AST; codegen
Builder	CST visitor → ASDL nodes
Helper API	<code>parse</code> , <code>cache</code> , <code>dump</code> , <code>walk</code> , <code>locations</code>
Unparser	Precedence-aware source regen
Visitor / transformer	Read vs rewrite
Type system	Qualifiers, UDT registry
Linter	Static rules, codes
Error model	<code>SyntaxError</code> , <code>indent</code> , <code>listener</code>

Internals

```
src/pynescrypt/ast/
  helper.py           # public pipeline + sha256 LRU parse cache
  builder.py          # PinescriptASTBuilder (_setLocations = direct attrs)
  unparser.py         # NodeUnparser + Precedence
  visitor.py          # NodeVisitor
  transformer.py       # NodeTransformer
  collector.py        # StatementCollector (annotation pairing)
  linter.py           # PineLinter
  type_system.py      # Type*, TypeRegistry, MethodResolver
  error.py            # SyntaxError / IndentationError
  node.py             # from grammar.asdl.generated import *
  grammar/
    antlr4/
      resource/        # EDIT HERE: .g4 + *Base.py
      generated/       # NEVER HAND-EDIT
      error_listener.py
      tool/generate.py
    asdl/
      resource/        # EDIT HERE: Pinescript.asdl
      generated/       # NEVER HAND-EDIT
      tool/generate.py
```

Thin wrappers `grammar/antlr4/lexer.py`, `parser.py`, `visitor.py` re-export generated classes so application code imports stable paths.

`parse` uses a thread-local ANTLR engine and a two-stage **SLL then LL** strategy; successful trees go into a sha256 LRU (`PYNE_PARSE_CACHE`, default 128).

Invariants

1. **Resource is source of truth.** Lexer/parser `.g4` and `Pinescript.asdl` define syntax and node shape; generated Python is a build product (committed so install does not require Java).
2. **Round-trip is a test oracle.** For the supported surface, `unparse(parse(src))` should re-parse and preserve semantics (not byte-identical formatting).
3. **Locations are first-class.** Statements and expressions carry `lineno`, `col_offset`, optional `end_*` for diagnostics and LSP.
4. **Annotations are not free-floating.** Comment tokens on `COMMENT_CHANNEL` are reified as `Comment` nodes and attached to `Script` / `FunctionDef` / `TypeDef` / `Assign` via kind suffixes (`S`, `F`, `T`, `V`).
5. **Recursion budget.** Deep ternary chains raise the Python recursion limit temporarily during parse (cap ≥ 5000).
6. **Cached trees are shared by identity.** Mutating a `parse()` result (transformer, `increment_lineno`) mutates the LRU entry. Call `clear_parse_cache()` after intentional mutation, or disable the cache.
7. **`FunctionDef.returns` is a type spec**, not an executable expression. Optional leading `type_specification` on UDFs/methods is stored there (0.3.9+).

Worked examples

Minimal script:



```
from pynescrypt.ast.helper import parse, dump

src = """
//@version=6
indicator("demo")
x = ta.sma(close, 14)
plot(x)
"""

tree = parse(src)
assert tree.__class__.__name__ == "Script"
assert any("version" in a for a in (tree.annotations or []))
print(dump(tree, include_attributes=True, indent=2))
```

Expression-only eval mode:

```
from pynescrypt.ast.helper import parse, literal_eval

expr = parse("1 + 2 * 3", mode="eval")
assert literal_eval(expr) == 7
```

Failure modes

Symptom	Likely cause	Where to look
<code>SyntaxError</code> with caret	Lexer/parser reject	Error model
<code>IndentationError</code>	Bad INDENT/DEDENT from LexerBase	<code>PinescriptLexerBase.py</code>
Missing <code>visit_*</code> after grammar change	Builder not updated for new rule	Builder
Hand-edit of <code>generated/</code> lost on regen	Violated resource-only rule	Grammar
Full regen breaks <code>builder.py</code>	Parser context accessors changed	Prefer targeted lexer refresh (v6 case study)

See also

- [Grammar \(ANTLR4\)](#)
- [Helper API](#)
- [Runtime](#)
- [LSP](#)
- [Compatibility](#)

GRAMMAR ANTLR4

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title: "Grammar (ANTLR4)" description: "Resource .g4 grammars, generated artifacts, INDENT/DEDENT, regeneration, and the never-edit-generated rule."

GRAMMAR (ANTLR4)

Concrete syntax for Pine Script in PYNE is defined by a split lexer/parser ANTLR4 grammar plus hand-written base classes that inject Python-like indentation tokens.

Abstract

Two grammars live under `src/pynescrypt/ast/grammar/antlr4/resource/`:

- `PinescriptLexer.g4` — keywords, operators, literals (including v6 triple-quoted strings), comments, hidden whitespace.



- `PinescriptParser.g4` — statements, expressions, type/enum/function/method declarations, control structures.

Python runtime classes under `generated/` are **outputs** of `antlr4 -Dlanguage=Python3`. They are committed so end users never need a JDK. **Never edit files under `generated/`** — changes evaporate on the next regeneration and fight the build pipeline.

Hand-written bases (copied into `generated/` by the generate tool):

- `resource/PinescriptLexerBase.py` — INDENT/DEDENT, newline elision, multiline string handling.
- `resource/PinescriptParserBase.py` — parser superClass hooks.

Conceptual model

```

flowchart LR
    subgraph resource ["resource/ - edit here"]
        LG4["PinescriptLexer.g4"]
        PG4["PinescriptParser.g4"]
        LB["PinescriptLexerBase.py"]
        PB["PinescriptParserBase.py"]
    end

    GEN["antlr4 tool"]
    subgraph generated ["generated/ - do not hand-edit"]
        LPY["PinescriptLexer.py"]
        PPY["PinescriptParser.py"]
        VIS["PinescriptParserVisitor.py"]
        LIS["PinescriptParserListener.py"]
        LBgen["*Base.py copies"]
    end

    WRAP["lexer.py / parser.py / visitor.py wrappers"]
    BLD["builder.py visit_*"]

    LG4 --> GEN
    PG4 --> GEN
    GEN --> LPY
    GEN --> PPY
    GEN --> VIS
    GEN --> LIS
    LB --> LBgen
    PB --> LBgen
    LPY --> WRAP
    PPY --> WRAP
    VIS --> BLD

```

Interface surface

Application code should not import `generated` symbols directly when a wrapper exists:

```

from pynescrypt.ast.grammar.antlr4.lexer import PinescriptLexer
from pynescrypt.ast.grammar.antlr4.parser import PinescriptParser
from pynescrypt.ast.grammar.antlr4.visitor import PinescriptParserVisitor
from pynescrypt.ast.grammar.antlr4.error_listener import PinescriptErrorListener

```



The public parse path is higher level — **Helper API** — which constructs lexer, token stream, parser, and strips default console error listeners in favor of `PinescriptErrorListener`.

Entry rules (`PinescriptParser.g4`)

Rule	Purpose
<code>start</code> → <code>start_script</code>	Full script (<code>statements?</code> <code>EOF</code>)
<code>start_expression</code>	Single expression + optional <code>NEWLINE</code> + <code>EOF</code>
<code>start_comments</code>	Comment stream only

Lexer structural facts

Token / channel	Behavior
<code>INDENT</code> / <code>DEDENT</code>	Imaginary tokens emitted by <code>PinescriptLexerBase</code> (declared in <code>tokens { }</code>)
<code>COMMENT</code>	<code>//</code> line comments → <code>COMMENT_CHANNEL</code> (not the default channel)
<code>HASH_COMMENT</code>	<code>#</code> ... line comments remapped → <code>type(COMMENT)</code> (must not swallow <code>#RRGGBB</code>)
<code>BACKTICKS</code>	Markdown fence <code>`</code> remapped to <code>COMMENT</code> (paste noise)
<code>WS</code>	Spaces/tabs/form-feed → <code>HIDDEN</code>
<code>UNICODE_NOISE</code>	Curly quotes, bullets, arrows → <code>HIDDEN</code>
<code>ERROR_TOKEN</code>	Catch-all <code>.</code> for recovery/reporting
<code>TRIPLE_DQ_STRING</code> / <code>TRIPLE_SQ_STRING</code>	v6 multiline strings, typed as <code>STRING</code>
<code>LSHIFT</code> / <code>RSHIFT</code> / <code>TILDE</code> / <code>AMP</code> / <code>PIPE</code> / <code>CARET</code>	Bitwise <code><<</code> <code>>></code> <code>~</code> <code>&</code> <code>`</code>

Keywords include `var`, `varip`, `method`, `export`, `enum`, `type`, qualifiers `const` / `input` / `simple` / `series`, and control forms `if` / `for` / `while` / `switch` / `once` (August 2026). Many of those words are **soft** in identifier position (`name` alternatives) so `as = input(...)` and `once = 3` parse.

Parser structural facts

- **Compound vs simple statements.** Functions, methods, types, enums, and structure-valued assignments are compound; imports, break/continue, expression statements, and simple assignments terminate with `NEWLINE`.
- **Trailing structures.** `a = 0, for i = 0 to n` is `trailing_structure_statements` (comma-separated simples ending in a structure).
- **Structures are dual-use.** `if` / `for` / `while` / `switch` appear both as statements and as **expressions** (`structure_expression`) — the classic Pine “if returns a value” form. **once is statement-only** (`once_structure` is not in `value_structure`); `x = once ...` is a parse error.
- **Local blocks.** Either indented (`NEWLINE` `INDENT` `statements` `DEDENT`) or inline single statement.
- **Library export.** `EXPORT?` prefixes function/method/type/enum and June-2025 `export const T name = ...` compound initializers.
- **UDF return types.** `function_declaration` / `method_declaration` accept an optional leading `type_specification` (`int ilog2(int n) => ...`). Builder stores it on `FunctionDef.returns`.



- **Bitwise layer.** Between `and` and comparisons: `|` `^` `&` then `<<` `>>`, then additive. Unary `~` is `Invert`.
- **Assign vs reassignment (0.3.9).** `=` on a bare name is **initialization** (`Assign` via `variable_declaration`). Reassignment of a name is `:=` (`ReAssign`). `=` is reassignment only for attribute / subscript targets (`obj.f =`, `a[i] =`).
- **Left-factored typed names.** `variable_declaration` is `declaration_mode? type_specification? name_store` with the typed alternative first so `int x = 1` does not consume `int` as the identifier. Same left-factor on `for_iterator` and parameter lists.

Internals

Paths

Path	Edit?	Role
<code>src/pynascript/ast/grammar/antlr4/resource/PinescriptLexer.g4</code>	Yes	Lexer grammar
<code>src/pynascript/ast/grammar/antlr4/resource/PinescriptParser.g4</code>	Yes	Parser grammar
<code>src/pynascript/ast/grammar/antlr4/resource/PinescriptLexerBase.py</code>	Yes	Indentation machine
<code>src/pynascript/ast/grammar/antlr4/resource/PinescriptParserBase.py</code>	Yes	Parser base
<code>src/pynascript/ast/grammar/antlr4/generated/**</code>	No	ANTLR output + copied bases
<code>src/pynascript/ast/grammar/antlr4/tool/generate.py</code>	Tool	Runs <code>antlr4</code> , copies <code>*.py</code> bases
<code>src/pynascript/ast/grammar/antlr4/error_listener.py</code>	Hand	Maps ANTLR errors → <code>ast.error.SyntaxError</code>

LexerBase responsibilities

`PinescriptLexerBase` is not optional sugar; it is part of the language definition:

- Ignore leading / collapse consecutive newlines; ensure trailing newline.
- Suppress newlines inside open `()` / `[]`, after operators, and for soft wrap lines whose indent width is not a multiple of four.
- Push `INDENT` / `DEDENT` with tab length and indent length = 4.
- Special-case multiline string tokens so wrap indentation inside strings is not mis-tokenized as structure.

Indent mistakes surface as `pynascript.ast.error.IndentationError` (subclass of the project `SyntaxError`).

Regeneration

```
# Project-aware entry (sets -o generated, -lib resource, copies *Base.py)
python -m pynascript.ast.grammar antlr4.tool.generate

# hatch lint:gen-parser is only the antlr4 CLI (`antlr4 {args}`) — pass flags yourself.
# hatch run lint:gen-parser -- -o ../generated -lib ../resource -listener -visitor -Dlanguage=Python3
```

`tool/generate.py` invokes `$(sys.executable)/../antlr4` with `-o generated -lib resource -listener -visitor -Dlanguage=Python3` on `resource/*.g4`, then copies `resource/*.py` into `generated/`. Requires a working `antlr4` CLI (hatch lint env pulls `antlr4-cli`). Generated modules are excluded from mypy/ruff strictness in `pyproject.toml` — do not “lint-fix” them by hand.



Downstream after grammar change

1. Regenerate (or selectively refresh lexer — see case study).
2. Add/adjust `visit_*` methods in `src/pynescrypt/ast/builder.py` for new or renamed rules.
3. If the **shape** of the language changed (new statement/expression kind), update `Pinescript.asdl` and regenerate ASDL nodes — [ASDL schema](#).
4. Smoke-test with `parse` / `unparse` on a minimal first-party snippet before broader regression.

Invariants

1. **Never hand-edit `generated/`**. Patches belong in `resource/*.g4` or `resource/*Base.py`.
2. **Visitor contract**. Generated `PinescriptParserVisitor` method names follow ANTLR rule names; the builder subclass must track renames.
3. **Comments are off the default channel**. Annotation logic in `helper._collect_comment_nodes` iterates `token_stream.tokens` for `COMMENT` type after `fill()`, and keeps only texts containing `@`. `HASH_COMMENT` / `BACKTICKS` are remapped to that same token type.
4. **Indent unit is four spaces**. Soft-wrap lines that are not multiples of four are newline-elided, not nested blocks.

Worked examples

Minimal grammar-level mental model

```
//@version=6
indicator("x")
if close > open
    plot(1)
else
    plot(0)
```

Lexer emits `NEWLINE`, `INDENT`, `DEDENT` around the `if` bodies. Parser builds `if_structure` → builder yields `ast.If` with `body` / `orelse` statement lists.

v6 triple-quoted strings

Resource lexer rules (illustrative — see live `PinescriptLexer.g4`):

```
TRIPLE_DQ_STRING : '"""' ( ~'"""' | '""" ~'"""' | '""" ~'"""' )* '"""' -> type(String) ;
TRIPLE_SQ_STRING : '\''\''\'' ( ~'\''\'' | '\'' ~'\''\'' | '\''\'' ~'\''\'' )* '\''\''\'' -> type(String) ;
```

ANTLR quoting pitfall: putting `"""` or `'''` as fragment literals can produce tool errors (`quote came as a complete surprise`). Factor starters when needed:

```
fragment TRIPLE_SQ_START: '\''\''\''\'';
```

Content and source indentation inside the triple quotes are preserved literally (no automatic dedent).



Typed names and `=` vs `:=` (0.3.9)

Pine lets type names occupy identifier positions (`int` is both a type and a legal `NAME` alternative via soft keywords). Alternatives must try the **typed** form first:

```
variable_declaration
  : declaration_mode? type_specification name_store
  | declaration_mode? name_store
  ;

simple_name_initialization: EXPORT? variable_declaration EQUAL expression;

simple_reassignment
  : assignment_target_attribute EQUAL expression
  | assignment_target_subscript EQUAL expression
  | primary_expression COLONEQUAL expression
  ;
```

Source	Node
<code>x = 1 / var float x = 1.0 / export const int N = 3</code>	<code>Assign (mode / type / export as present)</code>
<code>x := 2</code>	<code>ReAssign</code>
<code>obj.field = 3 / arr[i] = 4</code>	<code>ReAssign (attribute/subscript =)</code>
<code>x += 1</code>	<code>AugAssign</code>

Putting `name_store` before `type_specification name_store` made `int x = 1` parse `int` as the variable and then fail on `x`. Do not invert those alternatives.

Case study: targeted lexer refresh (2026-07)

Full regeneration once produced a `PinescriptParser.py` whose context accessors diverged from what the hand-written builder expected (e.g. missing `template_spec_suffix()`), breaking even trivial parses. The practical recovery:

1. Edit **only** `resource/` grammar.
2. Generate the lexer in a clean temp directory (avoids nested `src/` mirror paths).
3. Copy **only** `PinescriptLexer.py` (+ refreshed `LexerBase.py`) into `generated/`.
4. Leave committed `PinescriptParser.py` and visitors untouched unless prepared to patch the builder in the same change.
5. Verify immediately:

```
from pynescrypt.ast.helper import parse, unparse
ast = parse('indicator("x")\ns = ""\nfoo\n bar\n""\n')
assert "foo" in unparse(ast)
```



Failure modes

Failure	Cause	Mitigation
mismatched input after " " "	Old lexer ATN without triple rules	Refresh generated lexer from resource
Full regen breaks builder	Parser context API drift	Selective copy; or update all visit_* together
antlr4 not found	CLI not on PATH	Use hatch env or full path to antlr4-cli
Nested generated/src/pynescrypt/... junk	Running antlr with wrong -o cwd	Generate from clean temp + explicit -o
IndentationError mid-file	Mixed tabs/spaces or wrong nest	Align to 4-space blocks; check soft-wrap rules
Silent loss of hand patches	Edited generated/	Revert; re-apply in resource/

See also

- [ASDL schema](#)
- [Builder](#)
- [Error model](#)
- [Helper API](#)
- [Missing features](#)

ASDL SCHEMA

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title: "ASDL Schema" description: "Algebraic AST definition in Pinescript.asdl, generated dataclasses, and how schema changes flow into the pipeline."

ASDL SCHEMA

The abstract syntax tree is not free-form dicts. It is an **algebraic data type** declared in ASDL and materialized as Python dataclasses.

Abstract

`src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl` defines the `Pinescript` module: roots (`Script`, `Expression`), statements, expressions (including control structures that are also expressions), operators, params, args, cases, and comments. Codegen writes `grammar/asdl/generated/PinescriptASTNode.py`.



`src/pynescrypt/ast/node.py` re-exports that module so the rest of the codebase imports `from pynescrypt.ast import node as ast` (or `from pynescrypt.ast.node import Script`).

ASDL is the contract between **builder** (produces nodes), **evaluator/LSP** (consume nodes), and **unparser** (serializes nodes). Changing a field without regenerating and updating all three is a schema break.

Conceptual model



Each product sum type becomes a Python class hierarchy under `AST` with:

- `_fields` — logical children (walked by `iter_fields` / visitors)
- `_attributes` — location metadata (`lineno` , `col_offset` , `end_lineno` , `end_col_offset`) where declared

Interface surface

Module roots (`mod`)

Constructor	Fields	Use
<code>Script</code>	<code>stmt* body</code> , <code>string* annotations</code>	Full file (<code>mode="exec"</code>)
<code>Expression</code>	<code>expr body</code>	Single expression (<code>mode="eval"</code>)



Statements (stmt)

Constructor	Notable fields
FunctionDef	name , param* args , body , expr? returns , method? , export? , annotations
TypeDef	UDT type Name body (fields + methods as stmts)
EnumDef	enum Name members
Assign	target , optional value / type / mode (Var / VarIp) , export? , annotations
ReAssign	target := value , or obj.f = / a[i] = (not bare name =)
AugAssign	+= ... ops
Import	namespace/name/version , optional alias
Expr	Expression statement
Break / Continue	Loop control

All `stmt` and most other attributed products carry location attributes.

Expressions (expr)

Operators and primaries:

- `BoolOp` , `BinOp` , `UnaryOp` , `Conditional ( ? : )` , `Compare`
- `Call` , `Constant` , `Attribute` , `Subscript` , `Name` , `Tuple`
- Structures as expressions:** `ForTo` , `ForIn` , `While` , `If` , `Switch` — same shapes can appear as values (Pine’s expression-oriented control flow)
- Once(expr? test, stmt* body)** — August 2026 `once [cond]` block. Lives under `expr` like other structures so `structure_statement` can wrap it in `Expr` , but it is **not** a usable value (parser forbids it on `=` / `:=` RHS)
- `Qualify` — `const` / `input` / `simple` / `series` applied to a type/expression
- `Specialize` — generic specialization form (`value` + `args`)

Auxiliaries

Product	Role
decl_mode	Var VarIp
type_qual	Const Input Simple Series
expr_context	Load Store
param	Function/method parameter
arg	Call argument (optional keyword name)
case	Switch arm (pattern? , body)
cmnt / Comment	Comment with value and kind (annotation classification)

Operators are empty product constructors used as tags — the unparser maps class names back to tokens:

- Arithmetic / bool / compare: `Add` , `Sub` , `Mult` , `Div` , `Mod` , `And` , `Or` , `Eq` , `NotEq` , `Lt` , `LtE` , `Gt` , `GtE`
- Unary: `Not` , `UAdd` , `USub` , **`Invert`** (`~`)
- Bitwise (Pine v5+): **`BitAnd`** , **`BitOr`** , **`BitXor`** , **`LShift`** , **`RShift`**



Importing nodes

```
from pynescrypt.ast import node as ast

script = ast.Script(body=[
    ast.Expr(value=ast.Call(
        func=ast.Name(id="plot", ctx=ast.Load()),
        args=[ast.Arg(value=ast.Name(id="close", ctx=ast.Load()))],
    ))
])
```

Prefer constructing via `parse()` unless synthesizing trees for tests.

Internals

Paths

Path	Edit?
src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl	Yes — schema source of truth
src/pynescrypt/ast/grammar/asdl/generated/PinescriptASTNode.py	No — regenerated
src/pynescrypt/ast/grammar/asdl/tool/generate.py	Generator entry
src/pynescrypt/ast/node.py	Thin re-export (<code>from .grammar.asdl.generated import *</code>)

Generated class shape

Excerpt of generated form (do not edit this file; shown for orientation):

```
@dataclasses.dataclass
class FunctionDef(stmt):
    name: identifier = field(default=None)
    args: list[param] = field(default_factory=list)
    body: list[stmt] = field(default_factory=list)
    returns: expr | None = field(default=None)
    method: int | None = field(default=None)
    export: int | None = field(default=None)
    annotations: list[string] = field(default_factory=list)
    _fields: ClassVar[list[str]] = [
        "name", "args", "body", "returns", "method", "export", "annotations",
    ]
```

`stmt` base injects location `_attributes`. Hashability is forced to `object.__hash__` so nodes can live in sets/maps by identity when needed. `returns` is the optional leading type spec (`int f(x) => ...`), not a runtime return *value*.

Regeneration

When the schema changes:



```
# Project entry (wraps asdl/tool/asdlgen.py, then ruff format)
python -m pynescrypt.ast.grammar.asdl.tool.generate
# equivalent: asdlgen.py resource/Pinescript.asdl -o generated/PinescriptASTNode.py
```

Do not invoke bare `pyasdl ... -o generated/` — the project generator writes a **file** (`PinescriptASTNode.py`), not a directory of modules.

Then update, in the same change set:

1. `builder.py` construction sites
2. `unparser.py` `visit_*` methods
3. Evaluator expression/statement handlers
4. Any `isinstance` checks in LSP features

Dual nature of structures

ASDL places `If`, `ForTo`, `ForIn`, `While`, `Switch` under `expr`, while the builder often wraps structure-as-statement under `Expr(value=...)`. Annotation collection (`StatementCollector`) treats `assign/reassign/augassign/expr` that hold structure values specially so nested statements remain visible for `//@...` pairing.

Invariants

1. **Schema first.** New language constructs need an ASDL constructor (or a clear encoding into an existing one) before evaluator semantics.
2. **Fields vs attributes.** Visitors walk `_fields` only. Location is metadata; missing locations are fillable via `fix_missing_locations`.
3. **Optional flags as `int?`.** `method` and `export` are integer flags (truthy when set), not booleans — match builder assignments (`export = 1`).
4. **FunctionDef.returns is a type expression.** Engines must not `visit()` it as an executable value.
5. **Generated file is not a style playground.** Formatting/lint of `PinescriptASTNode.py` is overwritten on regen.

Worked examples

Map ASDL to a short script

```
//@version=6
f(x) => x + 1
a = f(close)
```

Approximate tree:

```
Script(
  annotations=["//@version=6", ...],
  body=[
    FunctionDef(name="f", args=[Param(name="x")], returns=None, body=[Expr(BinOp(...))]),
    Assign(target=Name("a", Store), value=Call(...)),
  ],
)
```



Dump fields programmatically

```
from pynescrypt.ast.helper import parse, iter_fields

tree = parse("x = 1")
for name, value in iter_fields(tree):
    print(name, type(value).__name__)
```

Failure modes

Failure	Cause
<code>AttributeError</code> on new field	Schema regenerated but consumer not updated
Builder constructs wrong product	ASDL and grammar out of sync
Unparser omits new node	Missing <code>visit_NewNode</code> → <code>generic_visit</code> may emit nothing useful
Hand-edit of generated nodes lost	Edited <code>generated/</code> instead of <code>.asdl</code>

See also

- [Builder](#)
- [Visitor / transformer](#)
- [Unparser](#)
- [Grammar \(ANTLR4\)](#)

BUILDER

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title: "AST Builder" description: "PinescriptASTBuilder: ANTLR parse-tree visitor that constructs ASDL nodes, locations, and comment kinds."

AST BUILDER

The builder is the semantic boundary between ANTLR's concrete parse tree and PYNE's ASDL abstract tree.

Abstract

`src/pynescrypt/ast/builder.py` defines three cooperating mixins:

1. **PinescriptASTLocator** — maps `ParserRuleContext.start/stop` tokens to `lineno` / `col_offset` / `end_*`.
2. **PinescriptCommentParser** — classifies `//...` text into annotation kinds (`@=S`, `@OF`, `#`, plain `//`, ...).



3. `PinescriptASTBuilder` — subclasses generated `PinescriptParserVisitor`, implements `visit_*` for every meaningful rule, returns ASDL nodes.

`helper._parse` constructs a builder, runs `builder.visit(rule_context)`, then (in exec mode) uses `StatementCollector` + comment tokens to attach annotations. The builder itself does not walk the default token channel for comments; that is a post-pass.

Conceptual model

```
sequenceDiagram
    participant H as helper._parse
    participant L as PinescriptLexer
    participant P as PinescriptParser
    participant B as PinescriptASTBuilder
    participant C as StatementCollector

    H->>L: tokenize stream
    H->>P: start_script / start_expression
    P-->>H: parse tree root ctx
    H->>B: visit(root)
    B-->>H: Script | Expression
    H->>C: visit(Script) -> stmts
    H->>B: _parseComment on COMMENT tokens
    H->>H: _add_annotations(script, stmts, comments)
```

Interface surface

You rarely instantiate the builder directly; `parse()` does. For tools that already hold a parse tree:

```
from pynescrypt.ast.builder import PinescriptASTBuilder
from pynescrypt.ast.grammar.antlr4.parser import PinescriptParser
# ... construct parser, parse to ctx ...
node = PinescriptASTBuilder().visit(ctx)
```

Locator API

The live path writes attributes **directly**. Do not describe `_setLocations` as a dict apply — that was an older micro-opt leftover.

Method	Role
<code>_setLocations(node, ctx)</code>	Hot path. Sets <code>node.lineno</code> / <code>col_offset</code> / <code>end_*</code> from <code>ctx.start</code> / <code>ctx.stop</code> (no intermediate dict).
<code>_getLocations(ctx)</code>	Unused helper that still returns a four-key dict. Not called by visit methods.

End column calculation accounts for embedded newlines in the stop token text. The builder instance is **stateless**; `helper._SHARED_BUILDER` reuses one object for every parse.

Comment kinds

`_parseComment` returns `(kind, parts)`:



Pattern	Kind prefix	Suffix	Example
<code>//@version = 5</code>	<code>@=</code>	S (script)	version assignment
<code>//@description ... / //@strategy_alert_message ...</code>	<code>@0</code>	S	script description / alert
<code>//@function / //@returns</code>	<code>@0</code>	F	function docs
<code>//@type</code>	<code>@0</code>	T	type docs
<code>//@variable</code>	<code>@0</code>	V	variable docs
<code>//@param name ...</code>	<code>@1</code>	F	param docs
<code>//@field name ...</code>	<code>@1</code>	T	field docs
<code>//# region / endregion</code>	<code>#</code>	—	editor regions
other <code>//</code>	<code>//</code>	—	ordinary comment

`helper._add_annotations` filters by suffix when attaching to nodes.

Store context

`_set_store_ctx` walks assignment targets (`Name` , nested `Tuple`) and sets `ctx = Store()` so load/store distinction is available to later passes (Python-ast style).

Representative visit methods

Rule visitor	Emits
<code>visitStart_script</code>	<code>Script(body)</code>
<code>visitStart_expression</code>	<code>Expression(body)</code>
<code>visitFunction_declaration / visitMethod_declaration</code>	<code>FunctionDef</code> (<code>method=1</code> for methods; optional <code>returns=</code> type spec)
<code>visitType_declaration</code>	<code>TypeDef</code>
<code>visitEnum_declaration</code>	<code>EnumDef</code>
<code>visitVariable_declaration</code>	<code>Assign</code> <code>shell(target / type / mode);</code> parent fills <code>value</code>
<code>visitSimple_name_initialization / visitCompound_name_initialization</code>	<code>Assign</code> (<code>export=1</code> if <code>EXPORT</code>)
<code>visitSimple_reassignment / visitCompound_reassignment</code>	<code>ReAssign</code> (<code>:=</code> , or <code>=</code> on attr/subscript)
<code>visit*_*_augassignment</code>	<code>AugAssign</code>
<code>visitIf_structure / visitFor_* / visitWhile_* / visitSwitch_* / visitOnce_structure</code>	Structure expressions (<code>once</code> is statement-only)
<code>visit*_*_expression</code> (disjunction $\rightarrow \dots \rightarrow$ primary)	Operator trees with correct associativity
<code>visitLiteral_*</code>	<code>Constant</code> (numbers via <code>ast.literal_eval</code> , strings, bools, colors)
<code>visitPrimary_expression_call</code>	<code>Call</code> + <code>Arg</code> list (positional/keyword)



Expression climbing follows the grammar's layered rules (`conditional` → `disjunction` → `conjunction` → **bitwise or/xor/and** → equality → inequality → **shift** → additive → multiplicative → unary (`~` / `not` / `+/-`) → primary).

`returns` is visited from `ctx.type_specification()` and must not be evaluated as a value by later engines.

Internals

Path

- Implementation: `src/pynescrypt/ast/builder.py`
- Visitor base: `src/pynescrypt/ast/grammar/antlr4/visitor.py` → generated `PinescriptParserVisitor`
- Node types: `src/pynescrypt/ast/node.py`
- Annotation pairing: `src/pynescrypt/ast/collector.py` + `helper._add_annotations`

Statement list flattening

`visitStatements` flattens each statement visitor's result: some rules return a **list** (simple multi-statement lines joined by commas), so the body is always a flat `list[stmt]`.

Export flag

When `EXPORT` is present on library declarations or name initialization, the builder sets `export = 1` on the corresponding node (integer flag per ASDL `int?`).

Assign vs ReAssign

`visitVariable_declaration` always builds `Assign` (optional `type` / `var` / `varip` mode). `=` after a bare or typed name is initialization. `visitSimple_reassignment` emits `ReAssign` for `obj.f =` / `a[i] =` / `name :=` — never for `name =`. See [Grammar](#).

Field / enum members

Field definitions become `Assign`-like statements inside `TypeDef.body` (with optional `VarIp` mode). Enum members become assignment-shaped stmts under `EnumDef.body`. The unparser special-cases method members of types when pretty-printing.

Invariants

1. **Every constructed stmt/expr that has a source span should receive `_setLocations`** (direct attribute writes) so LSP diagnostics and `get_source_segment` work.
2. **Visitor method names must track generated rule names.** Renaming a parser rule without updating the builder is a silent `generic_visit` fallback (wrong or empty trees).
3. **Do not edit `builder.py.bak`.** Live builder is `builder.py` only (stale backups are forbidden under project policy).
4. **Grammar and builder co-evolve.** A resource-only grammar change that adds rules is incomplete until `visit_*` exists.



Worked examples

What `x = close + 1` becomes

Rough structure after build:

```
Assign(  
  target=Name(id="x", ctx=Store()),  
  value=BinOp(  
    left=Name(id="close", ctx=Load()),  
    op=Add(),  
    right=Constant(value=1),  
  ),  
)
```

Export const initialization

Grammar: `EXPORT? variable_declaration EQUAL structure_expression`

Builder sets `assign.export = 1` when `ctx.EXPORT()` is present — required for library export of typed constants.

Typed UDF (`FunctionDef.returns`)

```
int ilog2(int n) =>  
  // ...
```

`visitFunction_declaration` stores the leading `type_specification` on `FunctionDef.returns` (here a `Name(id="int")`). Methods do the same with `method=1`. The unparser prints it back as a leading type.

Debugging a missing visitor

```
from pynescrypt.ast.helper import parse, dump  
print(dump(parse(your_snippet), indent=2))
```

If a subtree is `None` or a raw unexpected type, find the rule in `PinescriptParser.g4` and the matching `visitRule_name` in `builder.py`.

Failure modes

Symptom	Cause
<code>AttributeError: '...Context' object has no attribute '...'</code>	Full parser regen changed accessors; builder outdated
Locations all zero / missing	Forgot <code>_setLocations</code> on new visit method
Annotations not attached	Comment kind suffix wrong, or statement not yielded by <code>StatementCollector</code>
Tuples not assignable	Store context not propagated through <code>Tuple.elts</code>
Structure expression lost	Wrong rule branch between statement vs expression path



See also

- [Grammar \(ANTLR4\)](#)
- [ASDL schema](#)
- [Helper API](#)
- [Error model](#)

HELPER API

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title: "Helper API" description: "parse, sha256 LRU cache, unparse, dump, walk, literal_eval, and location utilities — the public AST surface."

HELPER API

`src/pynescrypt/ast/helper.py` is the stable library face of the language core. It orchestrates ANTLR, the builder, annotations, and the unparser behind functions deliberately similar to CPython's `ast` module.

Abstract

Call `parse` to obtain an ASDL tree; `dump` / `walk` / `iter_*` to inspect it; `unparse` to regenerate source; `literal_eval` for constant-folding-safe evaluation of literal expressions. Location helpers (`copy_location`, `fix_missing_locations`, `get_source_segment`, `increment_lineno`) support tooling that rewrites or reports on trees. `clear_parse_cache` / `parse_cache_info` control the process-local sha256 LRU.



Everything higher in the stack (CLI, LSP, Pro API, workers) ultimately routes through this module for front-end work.

Conceptual model

```
flowchart TB
    subgraph public ["Public API"]
        parse
        cache["clear_parse_cache / parse_cache_info"]
        unparse
        dump
        walk
        literal_eval
        locs["location helpers"]
    end

    subgraph private ["Private pipeline"]
        LRU["sha256 LRU (default ON)"]
        ENG["_ThreadParseEngine (TLS)"]
        _parse
        SLL["SLL then LL fallback"]
        BLD["_SHARED_BUILDER.visit"]
        COL["StatementCollector"]
        ANN["_add_annotations"]
    end

    parse --> LRU
    LRU --> ENG --> _parse
    _parse --> SLL --> BLD
    BLD --> COL --> ANN
    cache --> LRU
    unparse --> UP["unparse_node (TLS NodeUnparser)"]
    literal_eval --> NLE["NodeLiteralEvaluator"]
```

Interface surface

parse(source, filename="<unknown>", mode="exec") -> AST

Parameter	Meaning
source	Full script or expression text
filename	Error reporting name; resolved to absolute path if the file exists
mode	"exec" → start_script → Script ; "eval" → start_expression → Expression

Raises:

- `ValueError` — invalid mode
- `pynescrypt.ast.error.SyntaxError` — lexer/parser failure (via `PinescriptErrorListener`)

Side effects during exec mode:

1. Temporarily raises `sys.getrecursionlimit` to at least 5000.
2. Collects COMMENT tokens containing `@` and attaches annotations to eligible statements. If the source has no `@`, the annotation pass is skipped.

`filename` is **not** part of the cache key (only diagnostics).



Parse cache

Successful trees are stored in a process-local, thread-safe `OrderedDict` LRU keyed by `(sha256(source.encode("utf-8")), mode)`.

Control	Default	Meaning
<code>PYNE_PARSE_CACHE</code>	<code>on</code>	<code>0 / false / off / no</code> disables
<code>PYNE_PARSE_CACHE_MAX</code>	<code>128</code>	Max entries (must be ≥ 1)
<code>clear_parse_cache()</code>	—	Drop all entries and reset hit/miss counters
<code>parse_cache_info()</code>	—	<code>{enabled, size, maxsize, hits, misses}</code>

On a cache **hit**, the same AST object is returned (after scrubbing stale `_pine_call_site` attrs). Treat it as read-only: a `NodeTransformer` or `increment_lineno` mutates the cached entry for every later caller. After intentional mutation, call `clear_parse_cache()` or parse with the cache disabled.

`unparse(node) -> str`

Delegates to `unparse_node` — a **per-thread** reused `NodeUnparser` (warm visitor cache). Public return value is still a new `str`. See [Unparser](#).

`dump(node, *, annotate_fields=True, include_attributes=False, indent=None) -> str`

Recursive pretty-print of node trees. With `indent` (int spaces or string), multi-line layout is used for non-trivial nodes.

`literal_eval(node_or_string, context=None, data_feed=None, data_provider=None) -> Any`

- Strings are `parse(..., mode="eval")` first.
- Unwraps `Expression.body`.
- Uses `NodeLiteralEvaluator` — **not** full script evaluation. Non-literal graphs raise.

Tree navigation

Function	Behavior
<code>iter_fields(node)</code>	Yields <code>(name, value)</code> for set <code>_fields</code>
<code>iter_child_nodes(node)</code>	Direct AST children (incl. list items)
<code>walk(node)</code>	BFS over the full subtree

Location utilities

Function	Behavior
<code>copy_location(new, old)</code>	Copy <code>lineno/col/end_*</code> when both define the attribute
<code>fix_missing_locations(node)</code>	Fill gaps from parent defaults starting at <code>(1, 0)</code>
<code>increment_lineno(node, n=1)</code>	Shift all line numbers
<code>get_source_segment(source, node, *, padded=False)</code>	Slice original source by node span; <code>None</code> if incomplete ends



`__all__` export list

```
clear_parse_cache, copy_location, dump, fix_missing_locations, get_source_segment,
increment_lineno, iter_child_nodes, iter_fields, literal_eval, parse, parse_cache_info, unparse,
walk.
```

Internals

Path

`src/pynescrypt/ast/helper.py` — all public helpers above.

Related:

- `src/pynescrypt/ast/collector.py` — `StatementCollector` for annotation pairing
- `src/pynescrypt/ast/unparser.py` — `unparse_node` (thread-local `NodeUnparser`)
- `src/pynescrypt/ast/evaluator` — `NodeLiteralEvaluator` (`literal_eval` only)

Parse pipeline (`_parse`)

1. Validate `mode` $\in$ {`exec`, `eval`}.
2. Raise recursion limit to ≥ 5000 if needed.
3. Bind a **thread-local** lexer + token stream + parser (`_ThreadParseEngine`). Process-wide reuse is unsafe (indent / token-stream bleed).
4. **SLL first** (`BailErrorStrategy`). On `ParseCancellationException`, reset and re-parse **LL** (`DefaultErrorStrategy`). Trees match pure-LL on success.
5. `PinescriptASTBuilder.visit` via the shared stateless builder.
6. Exec mode: if source contains `@`, collect statements + `@` comments and `_add_annotations`.

Default ANTLR console listeners are removed; only `PinescriptErrorListener.INSTANCE` is installed.

Annotation algorithm (`_add_annotations`)

1. Merge comments and statements; sort by (`lineno`, `col_offset`).
2. Group consecutive comments vs statements.
3. Keep only kinds starting with `@`.
4. Script-level: first group members with kind ending in `S` $\rightarrow$ `script.annotations`.
5. Pair remaining comment groups with following statements; attach `F / T / V` filters to `FunctionDef` / `TypeDef` / `Assign`.

Comment **nodes** are not left in `Script.body`; only string annotation lists are stored on targets. Collection itself skips tokens whose text has no `@` (plain `//` and `//#` region never become `Comment` nodes).

Stream helpers

Internal	Role
<code>_parse_inputstream</code>	<code>InputStream(source)</code> + <code>stream.name = filename</code>
<code>_parse_filestream</code>	<code>FileStream</code> for on-disk scripts
<code>_get_absolute_path</code>	Resolve existing paths; leave <code><unknown></code> alone



Invariants

1. **mode** is a closed set. Only `exec` and `eval`.
2. **Default listeners are removed.** Console ANTLR spam is replaced by raising project `SyntaxError`.
3. **dump** requires an **AST** instance. Non-nodes raise `TypeError`.
4. **Round-trip tests should use `parse` + `unparse`**, not raw builder access, so annotation behavior matches production.
5. **Cache identity is shared.** Do not mutate a cached tree unless you also `clear_parse_cache()`.
6. **filename** is **diagnostic-only** and is not part of the sha256 key.

Worked examples

Inspect a tree

```
from pynescrypt.ast.helper import parse, dump, walk

tree = parse("""
//@version=6
indicator("demo")
plot(close)
""")
print(dump(tree, indent=2))
print(sum(1 for _ in walk(tree)), "nodes")
```

Source segment

```
from pynescrypt.ast.helper import parse, get_source_segment

src = "a = 1\nb = a + 2\n"
tree = parse(src)
assign_b = tree.body[1]
print(get_source_segment(src, assign_b)) # "b = a + 2\n" or similar span
```

Literal evaluation

```
from pynescrypt.ast.helper import literal_eval

assert literal_eval("2 * 3 + 4") == 10
assert literal_eval("'hi'") == "hi"
```



Parse cache

```
from pyonescript.ast.helper import parse, parse_cache_info, clear_parse_cache

src = 'indicator("x")\nplot(close)\n'
a = parse(src)
b = parse(src)
assert a is b # same object on hit
print(parse_cache_info()) # enabled, size, maxsize, hits, misses
clear_parse_cache()
```

Failure modes

Failure	Notes
Deeply nested ternaries hit recursion	Mitigated by temporary limit ≥ 5000 ; pathological depth can still fail
<code>get_source_segment</code> returns <code>None</code>	Missing <code>end_lineno</code> / <code>end_col_offset</code>
<code>literal_eval</code> raises	Non-literal AST (names, calls beyond allowed set)
Filename <code><unknown></code> in errors	Expected when parsing pure strings without a path
Annotations missing	Comment not <code>@...</code> form, or not immediately preceding eligible stmt
Transformer “leaks” across <code>parse()</code> calls	Cache hit returned the same object you mutated — clear or disable cache
<code>PYNE_PARSE_CACHE_MAX</code> ignored	Non-integer or <code>< 1</code> falls back to 128

See also

- [Builder](#)
- [Unparser](#)
- [Visitor / transformer](#)
- [Error model](#)

UNPARSER

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title: "Unparser" description: "NodeUnparser: precedence-aware AST → Pine Script source regeneration."

UNPARSER

The unparser is the inverse of the builder: ASDL nodes become source text with correct operator parenthesization and Pine idioms (`=>`, `:=`, `var / varip`, `export`, method vs function).

Abstract

`src/pynescrypt/ast/unparser.py` implements `NodeUnparser(NodeVisitor)` and a `Precedence` enum. `helper.unparse(node)` calls `unparse_node`, which reuses a **thread-local** `NodeUnparser` (warm type-visitor cache; `visit` still resets the buffer each call). Output is **semantically faithful**, not a pretty-printer of original trivia:



whitespace is normalized; comments that never became annotations are not reconstructed; annotation strings that *were* attached are re-emitted.

Round-trip tests (`parse` → `unparse` → `parse`) are the primary oracle for grammar/builder/unparser coherence.

Conceptual model

```
flowchart LR
    AST["AST tree"]
    V["visit / traverse"]
    PR["Precedence table"]
    BUF["_source list"]
    OUT["source string"]

    AST --> V
    PR --> V
    V --> BUF --> OUT
```

Parenthesization: higher `Precedence` values bind **tighter** (`TEST` loosest, `ATOM` tightest). A node is wrapped in `()` when the precedence **stored on it** (the parent context) is **strictly greater** than the node's own operator level. Equal levels do not wrap (left-associative chains).

Interface surface

```
from pynescrypt.ast.helper import parse, unparse

src = """
//@version=6
f(x) => x * 2
plot(f(close))
"""

print(unparse(parse(src)))
```

`Precedence` (low → high binding)

Matches `IntEnum` order in `unparser.py`:



Level	Operators / forms
TEST	ternary <code>?</code> <code>:</code> (weakest)
OR	<code>or</code>
AND	<code>and</code>
BITOR	<code> </code>
BITXOR	<code>^</code>
BITAND	<code>&</code>
EQ	<code>==</code> <code>!=</code>
INEQ / CMP	<code><</code> <code>></code> <code><=</code> <code>>=</code>
SHIFT	<code><<</code> <code>>></code>
EXPR	reserved slot (unused by current op tables)
ARITH	<code>+</code> <code>-</code>
TERM	<code>*</code> <code>/</code> <code>%</code>
FACTOR / NOT	unary <code>+</code> <code>-</code> <code>not</code> <code>~</code>
ATOM	names, literals, calls, subscripts

`Precedence.next()` steps one level tighter for right-hand associativity control. Bitwise tokens are emitted from `visit_BinOp` / `visit_UnaryOp` maps (`BitAnd` / `BitOr` / `BitXor` / `LShift` / `RShift` / `Invert`), not standalone `visit_BitAnd` methods.

Buffer helpers (internal but useful when subclassing)

Method	Role
<code>write(*text)</code>	Append fragments
<code>fill(text="")</code>	Newline + indent + text
<code>block()</code>	Indent context manager
<code>delimit(start, end)</code>	Wrap sub-output
<code>require_parens(prec, node)</code>	Conditional parentheses
<code>items_view</code> / <code>interleave</code>	Comma-separated lists
<code>buffered()</code>	Capture substring generation



Node coverage (high level)

Node	Emission sketch
Script	annotations then body
FunctionDef	optional <code>export / method</code> , optional <code>returns</code> type spec, <code>name(args) => body</code> (inline single <code>Expr</code> or indented block)
TypeDef	<code>type Name</code> with fields then methods
EnumDef	<code>enum Name</code> body
Assign	annotations, <code>export?</code> , <code>var / varip?</code> , <code>type?</code> , target <code>= value</code>
ReAssign	<code>target := value</code>
AugAssign	<code>target op= value</code>
ForTo / ForIn / While / If / Switch	Pine control syntax; <code>else if</code> chain collapse
Import	<code>import ns/name/version as alias?</code>
BinOp / BoolOp / UnaryOp / Compare / Conditional	precedence-aware, including bitwise <code>`</code>
Call / Attribute / Subscript / Name / Constant / Tuple	primaries

Internals

Path

- `src/pynescrypt/ast/unparser.py` — `Precedence`, `NodeUnparser`, `unparse_node`
- Entry: `helper.unparse` → `unparse_node` → thread-local `NodeUnparser.visit`

Visit vs traverse

`visit` resets `_source`, `_precedences`, and `_indent`, then calls `traverse`. `traverse` accepts lists (statement bodies) or single nodes, then dispatches via `_type_visitor_cache` (type-object keys, not class-name strings). This split lets statement lists and expression trees share code without double-buffering.

`visit_FunctionDef` emits `returns` (when set) as a leading type spec: `export method int name(args) => ...`.

If / else if chains

`visit_If` detects `orelse == [Expr(If(...))]` and emits `else if` rather than nested `else` + `if` blocks — matching idiomatic Pine.

`visit_Once` emits `once` plus an optional test and an indented body.

Type body ordering

`visit_TypeDef` partitions body into field statements vs `FunctionDef` with `method` set, emitting fields first then methods.



Constants

String constants use JSON-style quoting helpers where needed; colors and numbers re-serialize from their Python values. Exact original quote style (single vs double) is not guaranteed.

Invariants

1. **Precedence tables must stay aligned with the parser's expression layers** (including bitwise `|` `^` `&` `<<` `>>` between `and` and comparisons). A mismatch causes either redundant parens (benign) or wrong binding after re-parse (severe).
2. **Annotations are part of the round-trip surface** when attached to Script/FunctionDef/TypeDef/Assign.
3. **Unparser does not type-check.** Ill-typed but well-shaped trees still print.
4. **Indent unit is four spaces** (`" " * self._indent`).

Worked examples

Precedence

```
from pynescrypt.ast.helper import parse, unparse

# Builder produces BinOp(Add, left=1, right=BinOp(Mult, 2, 3)) or inverse depending on parse
print(unparse(parse("1 + 2 * 3", mode="eval")))
# Expect something that re-parses to the same value: e.g. "1 + 2 * 3"
```

Function forms

```
# Single-expression body → f(x) => expr
# Multi-statement body → f(x) =>\n    stmt\n    stmt
# With return type → int f(x) => expr
# Method → method int f(self) => expr
```

Reassignment vs declaration

```
// Assign with mode → var float x = 1.0
// ReAssign → x := 2.0
// AugAssign → x += 1
```



Failure modes

Symptom	Cause
Missing <code>visit_*</code> output	Falls through <code>generic_visit</code> which only walks children — may emit empty string for leaf-like unhandled nodes
Extra parentheses everywhere	Over-conservative precedence
Re-parse differs in binding	Under-parenthesized output
Lost comments	Only annotation strings are stored; plain <code>//</code> comments are dropped after parse
Method printed as function	<code>FunctionDef.method</code> flag not set by builder

See also

- [Helper API](#)
- [Builder](#)
- [ASDL schema](#)
- [Visitor / transformer](#)

VISITOR TRANSFORMER

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title: "Visitor & Transformer" description: "NodeVisitor and NodeTransformer: read-only traversal vs in-place AST rewrite with caching dispatch."

VISITOR & TRANSFORMER

Tree walks in PYNE follow the same dual pattern as CPython's `ast` module: a visitor for analysis, a transformer for structural rewrite.

Abstract

- **NodeVisitor** (`src/pynescrypt/ast/visitor.py`) — dispatch `visit_<ClassName>`, default `generic_visit` walks children via `iter_fields`. Method lookups are cached **per visitor instance, keyed by `type(node)`** (the type object, not the class-name string).



- **NodeTransformer** (`src/pynascript/ast/transformer.py`) — same dispatch, but `generic_visit` **replaces** list items and fields according to return values (`None` removes, list splices, AST replaces).

Specialized visitors elsewhere:

Class	Path	Role
<code>PinescriptASTBuilder</code>	<code>builder.py</code>	CST visitor (ANTLR), not <code>NodeVisitor</code>
<code>NodeUnparser</code>	<code>unparser.py</code>	Source emission
<code>StatementCollector</code>	<code>collector.py</code>	Yield statements for annotations
Evaluator classes	<code>evaluator/*.py</code>	Runtime interpretation

Conceptual model

```
flowchart TB
    N["AST node"]
    D["visit(node)"]
    C["_visitor_cache"]
    M["visit_ClassName or generic_visit"]

    N --> D
    D --> C
    C --> M

    subgraph visitor ["NodeVisitor.generic_visit"]
        V1["for field in iter_fields"]
        V2["visit children"]
        V3["discard return values"]
    end

    subgraph transformer ["NodeTransformer.generic_visit"]
        T1["for field in iter_fields"]
        T2["visit child"]
        T3["None → drop"]
        T4["list → extend"]
        T5["AST → replace"]
    end

    M --> visitor
    M --> transformer
```



Interface surface

NodeVisitor

```
from pyonescript.ast.visitor import NodeVisitor
from pyonescript.ast.helper import parse

class CallCounter(NodeVisitor):
    def __init__(self):
        super().__init__()
        self.count = 0

    def visit_Call(self, node):
        self.count += 1
        self.generic_visit(node)

tree = parse("plot(ta.sma(close, 14))")
c = CallCounter()
c.visit(tree)
assert c.count >= 2
```

Method	Contract
visit(node)	Resolve visit_<type(node).__name__> with cache; fallback generic_visit
generic_visit(node)	Recurse into AST fields and lists of AST

Returns are whatever the override returns (analysis often returns `None`).

NodeTransformer

```
from pyonescript.ast import node as ast
from pyonescript.ast.transformer import NodeTransformer
from pyonescript.ast.helper import parse, unparse

class RenameClose(NodeTransformer):
    def visit_Name(self, node: ast.Name):
        if node.id == "close":
            return ast.Name(id="open", ctx=node.ctx)
        return node

tree = parse("plot(close)")
tree = RenameClose().visit(tree)
assert "open" in unparse(tree)
```

Return value semantics when transforming a **list parent** (e.g. `body`):

Return	Effect
<code>None</code>	Remove this element
<code>AST</code>	Replace element
non-AST iterable	Splice multiple nodes in place of one
same node	Keep



For a **single AST field**, `None` deletes the attribute (`delattr`); otherwise `setattr` replaces.

StatementCollector

Generator-style visitor used only for annotation attachment:

- Yields `FunctionDef`, `TypeDef`, `EnumDef`, `Assign`, `ReAssign`, `AugAssign`, `Import`, `Expr`, `Break`, `Continue`.
- Descends into nested structures and into structure-valued assignments.
- Does **not** yield the structure nodes themselves as statements when they appear only as values — it yields their inner bodies.

```
from pynescrypt.ast.collector import StatementCollector
from pynescrypt.ast.helper import parse

stmts = list(StatementCollector().visit(parse("f() => 1\nx = 2")))
```

Internals

Paths

File	Symbols
<code>src/pynescrypt/ast/visitor.py</code>	<code>NodeVisitor</code>
<code>src/pynescrypt/ast/transformer.py</code>	<code>NodeTransformer</code>
<code>src/pynescrypt/ast/collector.py</code>	<code>StatementCollector</code> , <code>Structure</code> tuple
<code>src/pynescrypt/ast/helper.py</code>	<code>iter_fields</code> , <code>iter_child_nodes</code> , <code>walk</code>

Dispatch cache

```
cls = type(node)
visitor = cache.get(cls) or getattr(self, "visit_" + cls.__name__, generic_visit)
cache[cls] = visitor
```

`NodeUnparser` keeps a **second** type-keyed cache (`_type_visitor_cache`) used by `traverse`, bypassing `NodeVisitor.visit` so it does not reset the source buffer mid-render.

Subclass instances that dynamically add methods after first visit of a type will not see them unless the cache is cleared — prefer defining methods on the class body.

In-place lists

Transformers mutate list fields with `old_value[:] = new_values` rather than rebinding, so aliased references to the same list see updates.

Invariants

1. Always call `super().__init__()` on subclasses so `_visitor_cache` exists.
2. Transformers should return the node from overrides that use `generic_visit` (the default returns `node` after rewriting children).



- 3. **Do not assume identity stability** after a transformer pass if you replaced roots.
- 4. `walk` is not a visitor — it is a pure BFS helper without dispatch.

Worked examples

Strip all `Expr` statements that call `plot`

```
from pynescrypt.ast import node as ast
from pynescrypt.ast.transformer import NodeTransformer
from pynescrypt.ast.helper import parse, unparse

class DropPlots(NodeTransformer):
    def visit_Expr(self, node):
        if isinstance(node.value, ast.Call):
            func = node.value.func
            if isinstance(func, ast.Name) and func.id == "plot":
                return None
            return self.generic_visit(node)

tree = DropPlots().visit(parse("x = 1\nplot(x)\n"))
print(unparse(tree))
```

Collect all string constants

```
from pynescrypt.ast.visitor import NodeVisitor
from pynescrypt.ast.helper import parse

class Strings(NodeVisitor):
    def __init__(self):
        super().__init__()
        self.found = []

    def visit_Constant(self, node):
        if isinstance(node.value, str):
            self.found.append(node.value)

s = Strings()
s.visit(parse('indicator("hello")'))
```

Failure modes

Failure	Cause
<code>AttributeError: _visitor_cache</code>	Forgot <code>super().__init__()</code>
Transformer silently no-ops	Override returns nothing (<code>None</code>) unintentionally → node deleted
Infinite recursion	<code>visit_X</code> calls <code>self.visit(node)</code> on same node without change
Partial rewrite	Overrode <code>visit_*</code> without <code>generic_visit</code> — children untouched
Collector misses nested assign	Structure not in <code>Structure</code> tuple and not assigned as value



See also

- [Helper API](#)
- [Unparser](#)
- [Builder](#)
- [ASDL schema](#)

TYPE SYSTEM

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title: "Type System" description: "Pine v6 type model: qualifiers, builtins, collections, UDTs, TypeRegistry, and MethodResolver."

TYPE SYSTEM

PYNE's type layer models Pine's *qualified* types and user-defined types (UDTs) as Python objects usable by analysis and evaluation — not as a full independent typechecker IR (yet), but as the shared vocabulary for “what kind of value is this?”

Abstract

`src/pynescrypt/ast/type_system.py` defines:

- **Qualifiers** — `const`, `simple`, `series`, `input` (`TypeQualifier`)



- **Builtin kinds** — `int`, `float`, `bool`, `string`, `color`, `na`, `enum`
- **Composite types** — `array<T>`, `matrix<T>`, `map<K,V>`
- **UDTs** — `UserDefinedType` with `Field`s and `MethodSignature`s
- **Runtime instances** — `ObjectInstance` for UDT values
- **Registry / resolution** — `TypeRegistry`, `MethodResolver` (`.new`, `.copy`, user methods)

ASDL already has `Qualify` / `Specialize` expression forms and `type_qual` constructors; the type system module is the **semantic** counterpart used when evaluating UDT construction and when tools reason about types.

Conceptual model

```
classDiagram
  class Type {
    +name: str
    +qualifier: TypeQualifier?
    +is_compatible_with(other)
  }
  class BuiltinType {
    +kind: BuiltinTypeKind
  }
  class ArrayType {
    +element_type: Type
  }
  class MatrixType {
    +element_type: Type
  }
  class MapType {
    +key_type: Type
    +value_type: Type
  }
  class UserDefinedType {
    +fields: dict
    +methods: dict
    +is_exported: bool
  }
  Type <|-- BuiltinType
  Type <|-- ArrayType
  Type <|-- MatrixType
  Type <|-- MapType
  Type <|-- UserDefinedType
  UserDefinedType --> Field
  UserDefinedType --> MethodSignature
  ObjectInstance --> UserDefinedType
  TypeRegistry --> UserDefinedType
  TypeRegistry --> BuiltinType
  MethodResolver --> TypeRegistry
```



Interface surface

Qualifiers

```
from pynescrypt.ast.type_system import TypeQualifier

TypeQualifier.CONST    # compile-time constant
TypeQualifier.SIMPLE   # bar-invariant
TypeQualifier.SERIES   # per-bar series
TypeQualifier.INPUT    # input.* style parameter
```

Type.__str__ renders "series float" when a qualifier is set.

Builtins and factories

```
from pynescrypt.ast.type_system import (
    BuiltinTypeKind,
    int_type,
    float_type,
    bool_type,
    string_type,
    color_type,
    TypeQualifier,
)

t = float_type(TypeQualifier.SERIES) # "series float"
```

Factory	Kind
int_type	INT
float_type	FLOAT
bool_type	BOOL
string_type	STRING
color_type	COLOR

Collections

```
from pynescrypt.ast.type_system import ArrayType, MatrixType, MapType, int_type, string_type

ArrayType(int_type())          # array<int>
MatrixType(float_type())       # matrix<float>
MapType(string_type(), int_type()) # map<string, int>
```



UDTs

```
from pynescrypt.ast.type_system import UserDefinedType, Field, MethodSignature, float_type, int_type

point = UserDefinedType("point")
point.add_field(Field("x", float_type(), default_value=0.0))
point.add_field(Field("y", float_type(), default_value=0.0, varip=False))
point.add_method(MethodSignature("length", [], return_type=float_type()))
```

`Field` supports optional `default_value` and `varip` (mirrors `varip` fields in type declarations).

TypeRegistry

```
from pynescrypt.ast.type_system import TypeRegistry

reg = TypeRegistry()
reg.register_type(point)
assert reg.is_builtin_type("float")
assert reg.is_user_defined_type("point")
assert reg.get_type("float") is not None
```

Builtins are seeded in `_init_builtin_types` (`int`, `float`, `bool`, `string`, `color`, `na`, `enum`).

ObjectInstance + MethodResolver

```
from pynescrypt.ast.type_system import ObjectInstance, MethodResolver

inst = ObjectInstance(point)
inst.set_field("x", 1.5)
resolver = MethodResolver(reg)
# .new / .copy handled specially
copy = resolver.resolve_method(inst, "copy", [])
```

Method name	Behavior
<code>new</code>	Construct instance; positional args fill fields in declaration order
<code>copy</code>	Shallow copy of field dict
other	Lookup <code>UserDefinedType.methods</code> ; missing → <code>AttributeError</code>

Unknown field get/set on `ObjectInstance` raises `AttributeError` with type name context.

Relation to ASDL `type_qual`

ASDL	TypeQualifier
<code>Const</code>	<code>CONST</code>
<code>Input</code>	<code>INPUT</code>
<code>Simple</code>	<code>SIMPLE</code>
<code>Series</code>	<code>SERIES</code>

Builder produces `Qualify(qualifier=..., value=...)` for annotated types in the AST; the type system module is used when those need runtime/type-registry meaning (especially UDTs in builtins/evaluator).

Internals

Path

`src/pynescrypt/ast/type_system.py` — sole definition site for the classes above. **Not** star-exported from `pynescrypt.ast`; import the module directly.

Consumers (outside this doc's page set, for orientation):

- Evaluator UDT / collection code (`evaluator/builtins/`, `matrix_evaluator`, etc.)
- Future/static analysis hooks (LSP hover may grow toward this model)

Compatibility

`Type.is_compatible_with` is currently shallow: `BuiltinType` compares by equality; the base method returns `False` for other pairs. Treat richer subtyping (series/simple lattice, numeric promotion) as **not fully encoded** here — bar-loop semantics still own much of the coercion behavior.

na and enum

Registered as builtin kinds for name lookup. Enum *definitions* in scripts appear as AST `EnumDef`; linking enum members into `BuiltinTypeKind.ENUM` instances is evaluator-side work.

Invariants

1. **Registry lookup prefers builtins** over UDTs of the same name.
2. **Field presence is schema-true**. Setting an undeclared field is an error (no open bags).
3. **.new does not type-check arguments** against field types in the resolver — it assigns positionally.
4. **varip is metadata on Field**, not a separate type qualifier enum member.

Worked examples

Model a simple UDT from Pine

```
type candle
  float o
  float h
  float l
  float c
```



```
from pyonescript.ast.type_system import UserDefinedType, Field, float_type, TypeRegistry, MethodResc

candle = UserDefinedType("candle")
for name in "o h l c".split():
    candle.add_field(Field(name, float_type()))
reg = TypeRegistry()
reg.register_type(candle)
c = MethodResolver(reg).resolve_method(ObjectInstance(candle), "new", [1, 2, 0.5, 1.5])
assert c.get_field("h") == 2
```

Qualified series float

```
from pyonescript.ast.type_system import float_type, TypeQualifier
assert str(float_type(TypeQualifier.SERIES)) == "series float"
```

Failure modes

Failure	Cause
AttributeError: Field '...' not found	Typo or outdated UDT schema
AttributeError: Method '...' not found	Method never <code>add_method</code> 'd; not <code>new</code> / <code>copy</code>
Silent type confusion	<code>is_compatible_with</code> too weak — do not rely on it alone for safety
Name shadowing	UDT registered with a builtin name is unreachable via <code>get_type</code>

See also

- ASDL schema — `Qualify`, `TypeDef`, `EnumDef`
- Runtime — bar-loop use of values
- Builder — parsing type declarations
- Linters — style-level checks (not full typing)

LINTER

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title: "Linter" description: "PineLinter static rules: syntax via parse, version, deprecations, naming, style codes."

LINTER

The core linter is a lightweight static checker that sits on the parse pipeline plus regex rules. It is not a full dataflow analyzer; it is the first automated gate for “does this even look like modern Pine?”

Abstract

`src/pynescrypt/ast/linter.py` exports:

- `LintWarning` — dataclass (`code`, `message`, `line`, `column`, `severity`)
- `PineLinter` — stateful runner accumulating warnings



- `lint_script(source, filename)` / `lint_file(filepath)` — convenience entry points

Rules run in fixed order: **syntax** → **version** → **deprecated patterns** → **naming** → **style**. Syntax failures become severity `"error"` with code `E001`; other rules are mostly `"warning"`.

Conceptual model

```
flowchart TB
    SRC["source string"]
    L["PineLinter.lint"]
    S["_check_syntax → parse()"]
    V["_check_version"]
    D["_check_deprecated"]
    N["_check_naming"]
    STY["_check_style"]
    OUT["list of LintWarning"]

    SRC --> L --> S --> V --> D --> N --> STY --> OUT
```

Interface surface

```
from pynescrypt.ast.linter import lint_script, PineLinter, LintWarning

warnings = lint_script("""
indicator("x")
plot(close)
""")
for w in warnings:
    print(w) # warning: [W001] Missing @version ... at line 1
```

LintWarning

Field	Type	Meaning
<code>code</code>	<code>str</code>	Stable id (<code>E001</code> , <code>W001</code> , <code>C002</code> , ...)
<code>message</code>	<code>str</code>	Human-readable explanation
<code>line</code>	<code>int</code> <code>None</code>	1-based when known
<code>column</code>	<code>int</code> <code>None</code>	Reserved / optional
<code>severity</code>	<code>str</code>	<code>"warning"</code> (default) or <code>"error"</code>

`__str__` format: `{severity}: [{code}] {message} at {location}`.

`PineLinter.lint(source, filename=<input>)` -> `list[LintWarning]`

Resets `self.warnings`, runs all checks, returns the list (also stored on the instance).



File helper

```
from pynescrypt.ast.linter import lint_file

issues = lint_file("strategies/mean_reversion.pine")
```

Reads UTF-8 text then delegates to `lint_script`.

Rule catalog

Syntax

Code	Severity	Condition
E001	error	<code>parse(source, filename)</code> raises any exception

Does not attempt recovery; one syntax error check per run. Location is taken from `pynescrypt.ast.error.SyntaxError.details (lineno, offset)` when present; otherwise a `line N` regex on the exception text. The warning message prefers the short `.message` attribute so caret dumps do not land in chips.

Version

Code	Condition
W001	No <code>//@version = N</code> (flexible whitespace) match
W002	Version integer <code>< 5</code> (deprecated; suggest v6)

Pattern: `//@s*@version\s*=\s*(\d+)`.

Deprecated patterns (`_check_deprecated`)

Code	Pattern (simplified)	Advice
W101	<code>security('EXCHANGE:SYM'...) style</code>	Prefer <code>request.security()</code> with explicit params
W102	<i>(retired)</i> <code>plot(... style=plot.style_histogram</code>	Was: consider <code>plotcandle</code> . Code kept greppable; the rule no longer fires.
W103	<i>(retired)</i> <code>var int name = na</code>	Was: prefer <code>0</code> for type safety. Code kept greppable; the rule no longer fires.

Matches set `line` from prefix newline count. Case-insensitive search. Live deprecated rule is `W101` only.

Naming (`_check_naming`)

Code	Condition
C001	LHS of <code>name = ta...</code> starts with lowercase — message suggests camelCase via <code>_to_camel</code>

Heuristic only: line-local regex `(\w+)\s*=\s*ta\.`.



Style (`_check_style`)

Code	Condition
<code>C002</code>	Line length (rstrip) > 120
<code>C003</code>	(retired) Line matches <code>^\s+if\s+</code> — was “single-line if without braces”; code kept greppable
<code>C004</code>	File does not end with newline

Internals

Path

`src/pynescrypt/ast/linter.py`

Dependencies:

- `pynescrypt.ast.parse` (re-exported path via `from pynescrypt.ast import parse` — subject to the helper sha256 LRU)
- Standard library `re`, `dataclasses`

Design stance

The linter intentionally uses **regex over AST** for several rules so it still partially works when parse fails (version/deprecations/style still run after a failed syntax check — note: `_check_syntax` records `E001` but does not abort the pipeline). That means:

- False positives/negatives on fancy formatting are possible.
- Deep semantic issues (wrong series type, undefined names) belong to evaluator/LSP diagnostics, not these codes.

Integration points

CLI, editor save hooks, and CI can call `lint_script` without spinning up the full evaluator. LSP diagnostics may layer additional semantic checks beyond this module.

Invariants

1. `lint()` always clears prior warnings on that instance before running.
2. Codes are stable public strings — treat renames as breaking for tooling.
3. Syntax errors are not fatal to the function — you get `E001` plus any later regex hits.
4. No mutation of source — pure analysis.



Worked examples

Clean modern script

```
from pynescrypt.ast.linter import lint_script

src = """//@version=6
indicator("ok")
length = 14
basis = ta.sma(close, length)
plot(basis)
"""
assert lint_script(src) == [] # may still flag C001 if naming heuristic fires
```

Note: `basis = ta.sma(...)` triggers `C001` under the current rule (lowercase LHS). Prefer documenting that heuristic when teaching style.

Missing version

```
ws = lint_script('indicator("x")\nplot(close)\n')
assert any(w.code == "W001" for w in ws)
```

Programmatic filter

```
errors = [w for w in lint_script(src) if w.severity == "error"]
if errors:
    raise SystemExit(1)
```

Failure modes

Issue	Explanation
<code>C003</code> on legitimate indented <code>if</code> blocks	Rule retired in 0.6.0; the code no longer fires (was a coarse <code>^\s+if\s+</code> heuristic)
<code>C001</code> noise	Many valid snake_case or lower identifiers
<code>E001</code> still not a caret dump	Linters stores the short <code>.message</code> plus line/column; catch <code>pynescrypt.ast.error.SyntaxError</code> for the full caret <code>__str__</code>
Encoding errors in <code>lint_file</code>	Non-UTF-8 files raise at read time
False security deprecation	Pattern looks for quoted <code>EXCHANGE:SYM</code> form only

See also

- [Helper API](#) — underlying `parse`
- [Error model](#)
- [LSP diagnostics](#)
- [End-user troubleshooting](#)

ERROR MODEL

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title: "Error Model" description: "SyntaxError, IndentationError, SyntaxErrorDetails, and PinescriptErrorListener bridge from ANTLR to diagnostics."

ERROR MODEL

Parse failures in PYNE are not bare ANTLR console messages. They are structured exceptions with file, line, column, source excerpt, and a caret — suitable for CLI, tests, and LSP adapters.

Abstract

Two layers cooperate:

1. `src/pynescrypt/ast/error.py` — `SyntaxErrorDetails`, `SyntaxError`, `IndentationError`.



2. `src/pynascript/ast/grammar/antlr4/error_listener.py` — `PinescriptErrorListener` implements ANTLR's `ErrorListener.syntaxError`, builds details from the recognizer + offending token, and **raises** the project `SyntaxError`.

`helper._parse` installs this singleton listener on both lexer and parser after removing default listeners.

Indentation problems raised from `PinescriptLexerBase` use the same exception hierarchy (`IndentationError` subclass).

Note: these names intentionally shadow Python builtins (`SyntaxError`, `IndentationError`) inside the package — import carefully:

```
from pynascript.ast.error import SyntaxError as PineSyntaxError
```

Conceptual model

```
sequenceDiagram
    participant U as User code / parse()
    participant L as Lexer / Parser
    participant EL as PinescriptErrorListener
    participant E as SyntaxError

    U->>L: tokenize / parse
    L->>EL: syntaxError(recog, symbol, line, col, msg, exc)
    EL->>EL: filename, line text, end span from token
    EL->>E: raise SyntaxError(msg, details)
    E-->>U: exception with caret __str__
```

Interface surface

`SyntaxErrorDetails` (NamedTuple)

Field	Type	Meaning
<code>filename</code>	<code>str</code>	Path or <code><unknown></code>
<code>lineno</code>	<code>int</code>	1-based line
<code>offset</code>	<code>int</code>	0-based column
<code>text</code>	<code>str</code>	Source line (or excerpt)
<code>end_lineno</code>	<code>int</code> <code>None</code>	Multi-line span end
<code>end_offset</code>	<code>int</code> <code>None</code>	End column

`SyntaxError`(Exception)

```
class SyntaxError(Exception):
    def __init__(self, message: str, *details): ...
```

`details` may be:

- a single `SyntaxErrorDetails` instance, or
- positional components matching the NamedTuple fields.



Attributes: `message`, `details`.

`__str__` renders:

```
{message}
  File "{filename}", line {lineno}
    {stripped_line}
    {spaces}^
```

Offset is adjusted when the displayed line is `rstrip()`'d so the caret still points at the logical column.

IndentationError(SyntaxError)

Empty subclass for indent-stack failures from the lexer base (wrong nest, inconsistent levels). Catch either specifically or via the base `SyntaxError`.

PinescriptErrorListener

```
from pynescrypt.ast.grammarantlr4.error_listener import PinescriptErrorListener

listener = PinescriptErrorListener.INSTANCE # singleton
lexer.removeErrorListeners()
lexer.addErrorListener(listener)
parser.removeErrorListeners()
parser.addErrorListener(listener)
```

Method	Role
<code>syntaxError(...)</code>	Build details; raise <code>SyntaxError</code>
<code>_getFilenameFrom(recognizer)</code>	Walk Parser → TokenStream → Lexer → InputStream/FileStream
<code>_getInputTextFrom(recognizer, lineno)</code>	Full buffer or single line
<code>_splitLines</code>	Portable newline split (<code>\r\n</code> / <code>\n</code> / <code>\r</code>)

If the ANTLR exception `e` is already a project `SyntaxError`, its `details` are updated; otherwise the new error's `__cause__` is set to `e`.

Internals

Paths

Path	Role
<code>src/pynescrypt/ast/error.py</code>	Exception types
<code>src/pynescrypt/ast/grammar/antlr4/error_listener.py</code>	ANTLR bridge
<code>src/pynescrypt/ast/grammar/antlr4/resource/PinescriptLexerBase.py</code>	May raise <code>IndentationError</code> / <code>SyntaxError</code> during tokenization
<code>src/pynescrypt/ast/helper.py</code>	Wires listener; maps filename onto streams



Filename resolution order (InputStream)

1. `getSourceName()` if present
2. `sourceName` attribute
3. `input_stream.name` (set by helper to the user filename)
4. `"<unknown>"`

`FileStream` uses `fileName`.

End span from offending token

```
symbol_len = stop - start + 1
end_lineno = token.line + newlines_in_text
end_offset = adjusted if multiline else column + symbol_len
```

Mirrors the builder's location math so diagnostics and AST spans speak the same coordinate system.

Relationship to linter `E001`

`PineLinter._check_syntax` catches **any** `Exception` from `parse` and wraps it as `LintWarning(code="E001", severity="error")`. It **does** copy `details.lineno` / `details.offset` onto the warning when present, and prefers the short `.message` over the caret `__str__`. For the full caret rendering, catch `pynescrypt.ast.error.SyntaxException` at the call site instead of only reading linter output.

LSP / tooling

Adapters typically map:

Exception field	LSP Diagnostic
<code>lineno</code> / <code>offset</code>	<code>Range.start</code>
<code>end_lineno</code> / <code>end_offset</code>	<code>Range.end</code>
<code>message</code>	<code>message</code>
<code>IndentationError</code>	same, possibly different <code>code</code>

Invariants

1. **Default ANTLR listeners must stay removed** on the parse path — otherwise errors print twice and may not raise.
2. **Singleton listener is stateless** enough to share (`INSTANCE`); thread-local needs would require new instances.
3. **Raising aborts parse immediately** — no error recovery producing a partial tree through the public helper.
4. **Package `SyntaxError` ≠ builtin** — `except SyntaxError` without import may catch the wrong class.



Worked examples

Catching a parse error

```
from pynescrypt.ast.helper import parse
from pynescrypt.ast.error import SyntaxError as PineSyntaxError

try:
    parse("f( =>\n", filename="bad.pine")
except PineSyntaxError as e:
    print(e) # caret form
    print(e.details.lineno, e.details.offset)
    print(e.details.filename)
```

Distinguishing indentation

```
from pynescrypt.ast.error import IndentationError as PineIndentError
from pynescrypt.ast.error import SyntaxError as PineSyntaxError

try:
    parse("if true\nplot(1)\n") # missing indent under if – may indent-error depending on tokens
except PineIndentError as e:
    print("indent", e.message)
except PineSyntaxError as e:
    print("syntax", e.message)
```

Manual construction (tests)

```
from pynescrypt.ast.error import SyntaxError, SyntaxErrorDetails

details = SyntaxErrorDetails("t.pine", 3, 4, "    plot(\n", 3, 9)
err = SyntaxError("missing RPAR", details)
assert "t.pine" in str(err)
assert "^" in str(err)
```

Failure modes

Failure	Cause
<code>TypeError: unexpected type of input</code>	Listener received a recognizer/stream combo it does not support
Caret misaligned	Tabs vs spaces; display strips leading WS while offset counts raw
Filename always <code><unknown></code>	Stream name not set (bypassed helper)
Exception swallowed as generic <code>E001</code>	Only used linter path (line/column may still be filled from <code>details</code>)
<code>except SyntaxError</code> misses Pine errors	Caught builtin only
Partial trees	Not available via <code>parse()</code> after error — redesign would need recovery listeners

See also

- [Helper API](#)



- Grammar (ANTLR4)
- Linter
- LSP diagnostics
- Language core hub